

DMVPN

Dynamic Multipoint VPN

Scenario-Based Lab Implementation Guide

CCNP ENCOR 350-401 / SCOR 350-701

1. The Scenario — Why Are We Building This?

You are the network engineer for a company called NetCore Ltd. The company has one headquarters (HQ) and three branch offices. Management has approved a VPN solution to connect all locations securely over the internet.

Your manager hands you two options:

Option	Solution	Problem
A	Traditional GRE over IPsec (point-to-point)	Requires 3 separate tunnels on HQ. Branch-to-branch traffic must route through HQ. Every new branch = more manual work.
B	DMVPN	One tunnel interface on HQ handles all branches. Branches can talk directly to each other without going through HQ. New branches join automatically.

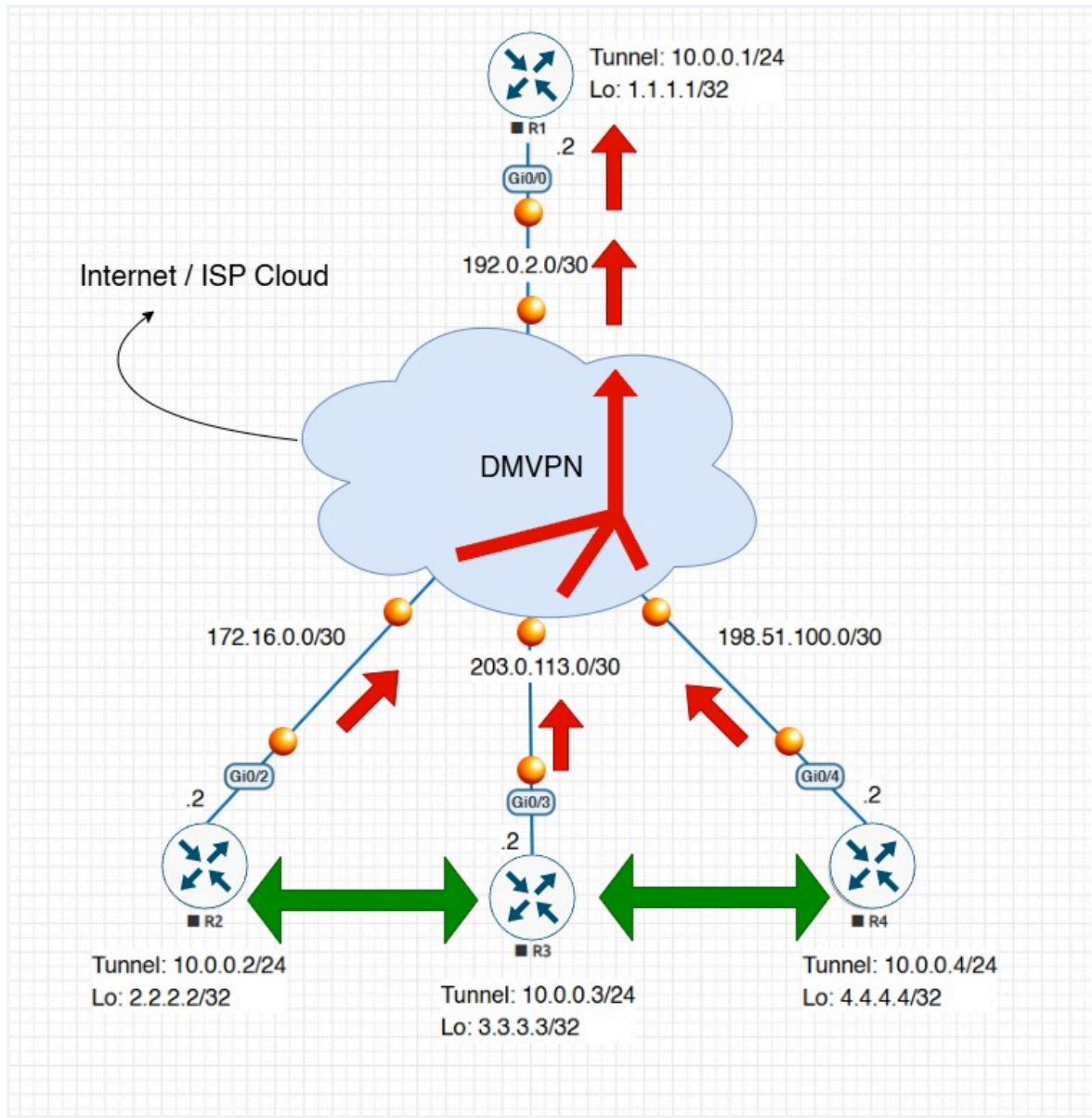
WHY: You choose DMVPN because it scales. With 3 branches today and 20 tomorrow, you cannot afford to manually configure individual tunnels for every pair of sites.

2. Lab Topology

The lab uses 4 routers representing the real-world scenario. R1 is HQ (the Hub). R2, R3, and R4 are branch offices (Spokes). The ISP cloud represents the public internet between them.

Device	Role	Tunnel IP	Physical WAN IP	Loopback
R1	Hub (HQ)	10.0.0.1 /24	192.0.2.2/30	1.1.1.1 /32
R2	Spoke 1 (Branch 1)	10.0.0.2 /24	172.16.0.1/30	2.2.2.2 /32
R3	Spoke 2 (Branch 2)	10.0.0.3 /24	203.0.113.1/30	3.3.3.3 /32
R4	Spoke 3 (Branch 3)	10.0.0.4 /24	198.51.100.1/30	4.4.4.4 /32
ISP	Internet	N/A	192.0.2.1/30	N/A

NOTE: The Tunnel IPs are all in the same subnet (10.0.0.0/24). This is the DMVPN overlay network — the virtual network that sits on top of the physical internet.



3. Understanding the Building Blocks Before You Configure

Before touching the CLI, you need to understand the three technologies inside DMVPN and why each exists. Skipping this step is why most engineers get confused during troubleshooting.

3.1 mGRE — Multipoint GRE

Regular GRE creates a point-to-point tunnel — one tunnel per destination. mGRE creates a single tunnel interface that can accept connections from multiple peers dynamically.

Regular GRE	mGRE
interface Tunnel0 → one destination only	interface Tunnel0 → accepts ANY peer dynamically
Need 3 tunnels for 3 branches	Need 1 tunnel interface for all branches
tunnel destination [specific IP]	tunnel mode gre multipoint — no fixed destination
Manual config for every new branch	New spokes join automatically

WHY: mGRE is what makes DMVPN scale. The Hub has one Tunnel interface regardless of how many spokes exist.

3.2 NHRP — Next Hop Resolution Protocol

NHRP is the phone book of DMVPN. Because spoke routers may have dynamic public IP addresses (DHCP from ISP), the Hub needs a way to know each spoke's real physical IP. NHRP solves this.

How NHRP works step by step:

1. Spoke R2 comes online and sends an NHRP Registration to the Hub
2. Hub records: Tunnel IP 10.0.0.2 = Physical IP 192.0.2.2
3. Spoke R3 does the same: Tunnel IP 10.0.0.3 = Physical IP 203.0.113.1
4. R2 wants to reach R3 directly — asks Hub: what is R3's real IP?
5. Hub responds: R3 is at 203.0.113.1
6. R2 builds a direct IPsec tunnel to 203.0.113.1 — bypasses Hub

NHRP Term	What It Means	Who Runs It
NHS — Next Hop Server	The Hub — holds the NHRP database	R1 (HQ)
NHC — Next Hop Client	The Spokes — register with NHS	R2, R3, R4
NHRP Registration	Spoke tells Hub its real IP on startup	Spokes send to Hub
NHRP Resolution	Spoke asks Hub for another spoke's real IP	Spoke-to-spoke queries
network-id	Identifies which DMVPN cloud this is — must match on all routers	All routers

WHY: NHRP is what allows spokes with dynamic IPs to join the DMVPN cloud automatically and find each other without manual config.

3.3 IPsec — Encryption

IPsec encrypts all DMVPN traffic. In DMVPN we use the IPsec Profile method (tunnel protection) — the same modern method from the GRE over IPsec lab. The profile is applied directly to the mGRE tunnel interface.

NOTE: *IPsec in DMVPN always uses Tunnel mode (not Transport mode) because the tunnel protection profile on an mGRE interface defaults to tunnel mode automatically.*

4. DMVPN Phases — Understanding Before Configuring

DMVPN has three phases. The phase determines whether spokes can talk directly to each other or must go through the Hub. This is a major exam topic.

Phase	Spoke-to-Spoke	Traffic Path	Key Difference
Phase 1	No	All traffic goes through Hub	Spokes use point-to-point GRE to Hub only. Hub is always in the data path.
Phase 2	Yes — direct tunnels	Spoke queries Hub via NHRP then builds direct tunnel	Spokes use mGRE. Direct tunnel persists until idle timeout.
Phase 3	Yes — more efficient	Hub sends NHRP redirect, spoke builds direct tunnel	More scalable than Phase 2. Requires NHRP redirect and shortcut on spokes.

EXAM TIP: *The phase is determined by the routing protocol configuration and NHRP settings — NOT by a single dmvpn phase command. Exams test whether you understand what changes between phases.*

In this lab we implement Phase 2 — the most commonly tested phase on ENCOR.

5. Step 1 — Base Configuration (Do This First)

Before building DMVPN, all routers need basic connectivity. Configure interfaces and verify physical reachability between all WAN interfaces. Without this, the DMVPN tunnel will never come up.

Config interfaces based on the topology (Interface gig , Loopback)

R1 —HUB 1 Base Config

```
R1(config)# hostname R1
R1(config)# interface Loopback0
R1(config-if)# ip address 1.1.1.1 255.255.255.255
R1(config)# interface GigabitEthernet0/0
R1(config-if)# ip address 192.0.2.2 255.255.255.252
R1(config-if)# no shutdown
```

R2 — Spoke 1 Base Config

```
R2(config)# hostname R2
R2(config)# interface Loopback0
R2(config-if)# ip address 2.2.2.2 255.255.255.255
R2(config)# interface GigabitEthernet0/2
R2(config-if)# ip address 172.16.0.2 255.255.255.252
R2(config-if)# no shutdown
```

R3 — Spoke 2 Base Config

```
R3(config)# hostname R3
R3(config)# interface Loopback0
R3(config-if)# ip address 3.3.3.3 255.255.255.255
R3(config)# interface GigabitEthernet0/3
R3(config-if)# ip address 203.0.113.2 255.255.255.252
R3(config-if)# no shutdown
```

R4 — Spoke 3 Base Config

```
R4(config)# hostname R4
R4(config)# interface Loopback0
R4(config-if)# ip address 4.4.4.4 255.255.255.255
R4(config)# interface GigabitEthernet0/4
R4(config-if)# ip address 198.51.100.2 255.255.255.252
R4(config-if)# no shutdown
```

ISP

```
Router(config)#hostname ISP
ISP(config)#interface GigabitEthernet0/0
ISP(config-if)#ip address 192.0.2.1 255.255.255.252
ISP(config-if)#no shutdown
ISP(config-if)#interface GigabitEthernet0/2
ISP(config-if)#ip address 172.16.0.1 255.255.255.252
ISP(config-if)#no shutdown
ISP(config-if)#interface GigabitEthernet0/3
ISP(config-if)#ip address 203.0.113.1 255.255.255.252
ISP(config-if)#no shutdown
```

```
ISP(config-if)#interface GigabitEthernet0/4
ISP(config-if)#ip address 198.51.100.1 255.255.255.252
ISP(config-if)#no shutdown
ISP(config-if)#end
```

Verify Base Connectivity:

```
R2# ping 192.0.2.1      ← R2 should reach R1
R3# ping 192.0.2.1      ← R3 should reach R1
R4# ping 192.0.2.1      ← R4 should reach R1
```

WHY: *If physical pings fail here, the DMVPN tunnel will never form. Always verify layer 3 reachability before building the overlay.*

6. Step 2 — Configure IPsec (Encryption Layer)

Configure IPsec on all routers before building the DMVPN tunnel. IPsec must be consistent across all devices — mismatched settings are the number one cause of DMVPN failures.

Configure on ALL Routers (R1, R2, R3, R4 — identical)

IKE Phase 1 Policy:

```
crypto isakmp policy 10
 encryption aes 256
 authentication pre-share
 group 14
 hash sha256
 lifetime 86400
```

WHY: *AES-256 and group 14 (2048-bit DH) are strong modern choices. Group 2 works but is considered weak. SHA-256 is preferred over MD5.*

Pre-Shared Key:

```
crypto isakmp key DMVPN-KEY address 0.0.0.0 0.0.0.0
```

WHY: *The wildcard address 0.0.0.0 0.0.0.0 means this key applies to ANY peer. This is required in DMVPN because spokes may have dynamic IPs — you cannot hardcode every peer address.*

IKE Phase 2 Transform Set:

```
crypto ipsec transform-set DMVPN-TS esp-aes 256 esp-sha256-hmac
mode tunnel
```

WHY: Tunnel mode is used here (unlike the GRE over IPsec lab which used transport). In DMVPN the IPsec profile is applied to the mGRE interface directly so tunnel mode is correct.

IPsec Profile:

```
crypto ipsec profile DMVPN-PROFILE
set transform-set DMVPN-TS
```

NOTE: The IPsec profile replaces the crypto map. It is applied directly to the tunnel interface using tunnel protection.

7. Step 3 — Configure the Hub Tunnel Interface (R1)

This is the most important configuration in DMVPN. The Hub tunnel interface is what makes DMVPN different from regular GRE over IPsec. Read every command and the WHY before typing it.

```
R1(config)# interface Tunnel0
ip address 10.0.0.1 255.255.255.0
```

WHY: This is the Hub's address in the DMVPN overlay network. All spokes will use this as their NHS (Next Hop Server) address.

```
tunnel source GigabitEthernet0/0
```

WHY: The physical interface the tunnel packets will originate from. This is the Hub's real public-facing IP (192.0.2.1). Spokes need to know this address.

```
tunnel mode gre multipoint
```

WHY: THIS is what makes it DMVPN instead of regular GRE. Multipoint means this single interface can accept tunnels from ANY spoke — not just one fixed destination. Without this command it is just regular point-to-point GRE.

```
ip nhrp network-id 1
```

WHY: Identifies which DMVPN cloud this interface belongs to. Must be the same number on ALL routers in the same DMVPN cloud. Think of it as a VLAN ID for the DMVPN overlay.


```
ip nhrp map multicast dynamic
```

WHY: Tells the Hub to dynamically learn which spokes should receive multicast traffic (like OSPF hellos). As each spoke registers via NHRP, the Hub adds them to the multicast replication list automatically. Without this, routing protocols cannot form adjacencies.

```
ip nhrp authentication NHRPAUTH
```

WHY: Optional but recommended. Prevents unauthorized routers from registering with the NHRP database. Must match on all routers.

```
tunnel protection ipsec profile DMVPN-PROFILE
```

WHY: Applies IPsec encryption to everything passing through this tunnel interface. Same tunnel protection method from your GRE over IPsec lab — now applied to mGRE.

```
ip mtu 1400  
ip tcp adjust-mss 1360
```

WHY: GRE adds 24 bytes of overhead. IPsec adds more. Without reducing the MTU, large packets will be fragmented or dropped silently. This prevents hard-to-troubleshoot connectivity issues where small pings work but large transfers fail.

Complete Hub Tunnel Configuration:

```
interface Tunnel0  
ip address 10.0.0.1 255.255.255.0  
tunnel source GigabitEthernet0/0  
tunnel mode gre multipoint  
ip nhrp network-id 1  
ip nhrp map multicast dynamic  
ip nhrp authentication NHRP-AUTH  
tunnel protection ipsec profile DMVPN-PROFILE  
ip mtu 1400  
ip tcp adjust-mss 1360
```

8. Step 4 — Configure Spoke Tunnel Interfaces (R2, R3, R4)

Spoke configuration is similar to the Hub but with key differences. Spokes must point back to the Hub as their NHS and map the Hub's tunnel IP to its physical IP.

R2 — Spoke 1 Tunnel Configuration

```
interface Tunnel0
ip address 10.0.0.2 255.255.255.0
```

WHY: Each spoke gets a unique IP in the same overlay subnet as the Hub (10.0.0.0/24).

```
tunnel source GigabitEthernet0/2
```

WHY: R2's physical WAN interface (192.0.2.2). This is what the Hub and other spokes will use to reach R2.

```
tunnel mode gre multipoint
```

WHY: Spokes also need multipoint mode for Phase 2. This allows the spoke to build direct tunnels to other spokes without going through the Hub. In Phase 1 you would use regular tunnel mode here.

```
ip nhrp network-id 1
```

WHY: Must match the Hub's network-id exactly. This ties R2 to the same DMVPN cloud as R1.

```
ip nhrp nhs 10.0.0.1
```

WHY: NHS = Next Hop Server = the Hub. This tells R2 where to send its NHRP registration and where to query for other spokes' real IPs. This is the Hub's TUNNEL IP not its physical IP.

```
ip nhrp map 10.0.0.1 192.0.2.2
```

WHY: This is the static NHRP mapping that bootstraps the process. R2 needs to know: Hub tunnel IP 10.0.0.1 = Hub physical IP 192.0.2.1. Without this R2 cannot send the initial NHRP registration because it does not know how to reach the Hub.

```
ip nhrp map multicast 192.0.2.2
```

WHY: Tells R2 to send multicast packets (OSPF hellos, EIGRP updates) to the Hub's physical IP. The Hub then replicates them to all other spokes via the dynamic multicast map configured on the Hub.

```
ip nhrp authentication NHRPAUTH
tunnel protection ipsec profile DMVPN-PROFILE
ip mtu 1400
ip tcp adjust-mss 1360
```

R3 — Spoke 2 (only difference is IP addresses)

```
interface Tunnel0
ip address 10.0.0.3 255.255.255.0
no ip redirects
ip mtu 1400
ip nhrp authentication NHRPAUTH
ip nhrp map 10.0.0.1 192.0.2.2
ip nhrp map multicast 192.0.2.2
ip nhrp network-id 1
ip nhrp nhs 10.0.0.1
ip tcp adjust-mss 1360
tunnel source GigabitEthernet0/3
tunnel mode gre multipoint
tunnel protection ipsec profile DMVPN-PROFILE
```

R4 — Spoke 3

```
interface Tunnel0
ip address 10.0.0.4 255.255.255.0
no ip redirects
ip mtu 1400
ip nhrp authentication NHRPAUTH
ip nhrp map 10.0.0.1 192.0.2.2
ip nhrp map multicast 192.0.2.2
ip nhrp network-id 1
ip nhrp nhs 10.0.0.1
ip tcp adjust-mss 1360
tunnel source GigabitEthernet0/4
tunnel mode gre multipoint
tunnel protection ipsec profile DMVPN-PROFILE
```

9. Step 5 — Configure Routing (Underlay and Overlay)

For each router to have connectivity we have to config bgp peering with each other. ISP advertises default route 0.0.0.0/0 to all BGP neighbors via default-originate. Every spoke and hub automatically knows how to reach the internet and each other's public WAN IPs through the ISP

ISP-BGP

```
router bgp 65000
 neighbor 192.0.2.2 remote-as 65001
 neighbor 203.0.113.2 remote-as 65003
 neighbor 198.51.100.2 remote-as 65004
 neighbor 172.16.0.2 remote-as 65002

 neighbor 192.0.2.2 default-originate
 neighbor 203.0.113.2 default-originate
 neighbor 198.51.100.2 default-originate
 neighbor 172.16.0.2 default-originate
```

R1, R2, R3, R4 – BGP

```
router bgp 65001
 neighbor 192.0.2.1 remote-as 65000
```

DMVPN needs a routing protocol to advertise networks across the overlay. EIGRP is the most common choice for DMVPN Phase 2 because of how it handles split-horizon.

NOTE: *Split-horizon is a critical concept here. By default, EIGRP will not advertise routes back out the same interface they were learned on. On the Hub, this means spoke routes learned on Tunnel0 will NOT be advertised to other spokes — breaking spoke-to-spoke routing. You must disable split-horizon on the Hub.*

R1 — Hub EIGRP Configuration

```
router eigrp 100
 network 10.0.0.0 0.0.0.255
 network 1.1.1.1 0.0.0.0
 no auto-summary

interface Tunnel0
 no ip split-horizon eigrp 100
```

WHY: *Disabling split-horizon on the Hub tunnel interface allows the Hub to re-advertise spoke routes back out Tunnel0 to other spokes. Without this, Branch1 cannot learn how to reach Branch2 through the DMVPN.*

R2, R3, R4 — Spoke EIGRP Configuration (same on all)

```
router eigrp 100
 network 10.0.0.0 0.0.0.255
 network 2.2.2.2 0.0.0.0
 no auto-summary
```

NOTE: *Spokes do NOT need split-horizon disabled because they only have one tunnel peer (the Hub initially). The Hub handles route redistribution between spokes.*

Split-Horizon Rule	Hub	Spoke
Disable split-horizon on Tunnel0?	YES — required for spoke-to-spoke routing	NO — not needed
Why?	Hub learns spoke routes on Tunnel0 and must re-advertise them back out Tunnel0 to other spokes	Spokes only send routes to the Hub

10. Step 6 — Verification (Do This in Order)

Always verify in layers. Start at the bottom (IPsec/NHRP) and work up (routing/connectivity). Jumping straight to ping when things fail wastes time.

Layer 1 — Verify IPsec

```
R1# show crypto isakmp sa
Expected: QM_IDLE ACTIVE for each spoke
```

```
R1# show crypto ipsec sa
Expected: pkts encaps/encrypt incrementing
```

WHY: If IKE Phase 1 is not ACTIVE, check pre-shared keys and ISAKMP policy match. If Phase 2 is not up, check transform-set consistency across all routers.

Layer 2 — Verify NHRP

```
R1# show ip nhrp
Expected: Each spoke's tunnel IP mapped to its physical IP
Example: 10.0.0.2/32 via 10.0.0.2 Tunnel0
         192.0.2.2 (registered)
```

```
R2# show ip nhrp
Expected: Hub mapping + any resolved spoke mappings
```

WHY: If spokes are not registering, check network-id matches, NHS IP is correct, and the static NHRP map points to the Hub's correct physical IP.

Layer 3 — Verify Tunnel Interface

```
R1# show interface Tunnel0
Expected: Tunnel0 is up, line protocol is up
```

```
R1# show ip interface brief
Expected: Tunnel0 10.0.0.1 YES up up
```

Layer 4 — Verify Routing

```
R1# show ip eigrp neighbors
Expected: R2, R3, R4 all showing as EIGRP neighbors
```

```
R2# show ip route eigrp
Expected: Routes to 3.3.3.3, 4.4.4.4 via 10.0.0.1 (Hub)
```

WHY: If EIGRP neighbors are not forming, verify multicast reachability through the tunnel. Check that `ip nhrp map multicast` is configured correctly on all spokes and `ip nhrp map multicast dynamic` is on the Hub.

Layer 5 — Verify Spoke-to-Spoke (Phase 2)

```
R2# ping 3.3.3.3 source loopback 0
First ping may fail or be slow — NHRP resolution in progress
Second ping should succeed via direct spoke-to-spoke tunnel
```

```
R2# show ip nhrp
After pinging R3, you should see R3's entry marked as 'dynamic'
10.0.0.3 via 10.0.0.3 Tunnel0 dynamic (spoke-to-spoke resolved)
```

WHY: The first ping triggers NHRP resolution. R2 asks Hub for R3's real IP. Hub responds. R2 builds a direct tunnel. Subsequent pings go directly R2 to R3 without touching the Hub.

Complete Verification Command Reference

Command	Run On	What to Look For
show crypto isakmp sa	All	QM_IDLE ACTIVE for all peers
show crypto ipsec sa	All	pkts encrypt/decrypt incrementing
show ip nhrp	Hub	All spokes registered with tunnel-to-physical mapping
show ip nhrp	Spoke	Hub static map + dynamic spoke entries after resolution
show dmvpn	All	Shows NHRP state, peers, tunnel status in one view
show ip eigrp neighbors	All	All expected neighbors present
show ip route eigrp	Spokes	Routes to all other loopbacks via DMVPN
show interface Tunnel0	All	Up/Up status
debug ip nhrp	Troubleshoot	Shows NHRP registration and resolution messages
debug crypto isakmp	Troubleshoot	Shows Phase 1 negotiation — find mismatch

11. Troubleshooting Guide — What Went Wrong?

Symptom	Most Likely Cause	Fix
Tunnel interface is down/down	IPsec not configured or Phase 1 failing	Check crypto isakmp sa — verify keys match on both sides
Tunnel up but no NHRP registrations	Spokes cannot reach Hub physically	Verify physical ping from spoke WAN IP to Hub WAN IP
NHRP registered but no EIGRP neighbors	Multicast not flowing through tunnel	Check ip nhrp map multicast on spokes and ip nhrp map multicast dynamic on Hub
Spoke routes not in routing table	Split-horizon blocking re-advertisement	Add no ip split-horizon eigrp 100 on Hub Tunnel0
Spoke-to-spoke Phase 2 not working	Spokes using point-to-point GRE not mGRE	Verify tunnel mode gre multipoint on spokes
Large transfers fail but pings work	MTU mismatch — GRE+IPsec overhead	Add ip mtu 1400 and ip tcp adjust-mss 1360 on all tunnel interfaces
NHRP authentication fails	Mismatched NHRP auth string	Verify ip nhrp authentication string matches on all routers
First spoke-to-spoke ping fails	Normal — NHRP resolution takes time	Send second ping — it should succeed after NHRP resolves

12. Phase 1 vs Phase 2 — Configuration Differences

Now that you have built Phase 2, here is exactly what changes between phases. This is critical exam knowledge.

Configuration	Phase 1	Phase 2
Spoke tunnel mode	tunnel mode gre (point-to-point)	tunnel mode gre multipoint
Hub split-horizon	Not needed (no spoke-to-spoke)	Must disable: no ip split-horizon
Spoke-to-spoke traffic	Always through Hub	Direct after NHRP resolution
NHRP on spokes	Basic registration only	Full registration + resolution queries
Routing protocol next-hop	Hub is always next-hop	Spoke becomes next-hop after resolution (Phase 2 specific)

EXAM TIP: On ENCOR exam: Phase 2 requires next-hop-self to be DISABLED on OSPF or EIGRP on the Hub so spokes can learn the correct next-hop for spoke-to-spoke traffic. If

| *next-hop-self is enabled all routes point to the Hub and spoke-to-spoke direct tunnels never form.*

13. Memory Hooks — Lock These In

| **DMVPN = mGRE + NHRP + IPsec**

Remove any one of these three and DMVPN breaks. mGRE = scalability. NHRP = the phone book. IPsec = the lock.

| **NHRP = PIM RP (The same theory from multicast)**

Spokes register with Hub like sources register with RP. Spokes query Hub for peer IPs like receivers query RP for source. After resolution both RP and Hub are out of the data path.

| **Hub split-horizon = always off in Phase 2**

If spokes cannot learn each other's routes check split-horizon first. It is the most common Phase 2 routing failure.

| **Static NHRP map bootstraps the process**

ip nhrp map [hub-tunnel-ip] [hub-physical-ip] on every spoke is required. Without it spokes cannot send their initial registration because they do not know how to reach the Hub.

14. Full Configuration

ISP Configurations

```
hostname ISP
!
!
interface GigabitEthernet0/0
ip address 192.0.2.1 255.255.255.252
duplex full
speed 1000
media-type rj45
!
interface GigabitEthernet0/2
ip address 172.16.0.1 255.255.255.252
duplex full
speed 1000
media-type rj45
!
interface GigabitEthernet0/3
ip address 203.0.113.1 255.255.255.252
duplex full
speed 1000
media-type rj45
!
interface GigabitEthernet0/4
ip address 198.51.100.1 255.255.255.252
duplex full
speed 1000
media-type rj45
!
router bgp 65000
bgp log-neighbor-changes
neighbor 172.16.0.2 remote-as 65002
neighbor 172.16.0.2 default-originate
neighbor 192.0.2.2 remote-as 65001
neighbor 192.0.2.2 default-originate
neighbor 198.51.100.2 remote-as 65004
neighbor 198.51.100.2 default-originate
neighbor 203.0.113.2 remote-as 65003
neighbor 203.0.113.2 default-originate
!
```

R1 Configurations

```
hostname R1
!
!
crypto isakmp policy 10
  encr aes 256
  hash sha256
  authentication pre-share
  group 14
crypto isakmp key DMVPN-KEY address 0.0.0.0
!
!
crypto ipsec transform-set DMVPN-TS esp-aes 256 esp-sha256-hmac
mode tunnel
!
crypto ipsec profile DMVPN-PROFILE
set transform-set DMVPN-TS
!
!
interface Loopback0
ip address 1.1.1.1 255.255.255.255
!
interface Tunnel0
ip address 10.0.0.1 255.255.255.0
no ip redirects
ip mtu 1400
no ip split-horizon eigrp 100
ip nhrp authentication NHRPAUTH
ip nhrp map multicast dynamic
ip nhrp network-id 1
ip tcp adjust-mss 1360
tunnel source GigabitEthernet0/0
tunnel mode gre multipoint
tunnel protection ipsec profile DMVPN-PROFILE
!
interface GigabitEthernet0/0
ip address 192.0.2.2 255.255.255.252
duplex full
speed 1000
media-type rj45
!
!
router eigrp 100
network 1.1.1.1 0.0.0.0
network 10.0.0.0 0.0.0.255
!
router bgp 65001
bgp log-neighbor-changes
neighbor 192.0.2.1 remote-as 65000
!
```

R2 Configurations

```
hostname R2
!
crypto isakmp policy 10
  encr aes 256
  hash sha256
  authentication pre-share
  group 14
crypto isakmp key DMVPN-KEY address 0.0.0.0
!
!
crypto ipsec transform-set DMVPN-TS esp-aes 256 esp-sha256-hmac
mode tunnel
!
crypto ipsec profile DMVPN-PROFILE
  set transform-set DMVPN-TS
!
!
interface Loopback0
  ip address 2.2.2.2 255.255.255.255
!
interface Tunnel0
  ip address 10.0.0.2 255.255.255.0
  no ip redirects
  ip mtu 1400
  ip nhrp authentication NHRPAUTH
  ip nhrp map 10.0.0.1 192.0.2.2
  ip nhrp map multicast 192.0.2.2
  ip nhrp network-id 1
  ip nhrp nhs 10.0.0.1
  ip tcp adjust-mss 1360
  tunnel source GigabitEthernet0/2
  tunnel mode gre multipoint
  tunnel protection ipsec profile DMVPN-PROFILE
!
interface GigabitEthernet0/2
  ip address 172.16.0.2 255.255.255.252
  duplex auto
  speed auto
  media-type rj45
!
!
router eigrp 100
  network 2.2.2.2 0.0.0.0
  network 10.0.0.0 0.0.0.255
!
router bgp 65002
  bgp log-neighbor-changes
  neighbor 172.16.0.1 remote-as 65000
  neighbor 203.0.113.1 remote-as 65000
```

R3 Configurations

```
hostname R3
!
crypto isakmp policy 10
  encr aes 256
  hash sha256
  authentication pre-share
  group 14
crypto isakmp key DMVPN-KEY address 0.0.0.0
!
!
crypto ipsec transform-set DMVPN-TS esp-aes 256 esp-sha256-hmac
mode tunnel
!
crypto ipsec profile DMVPN-PROFILE
set transform-set DMVPN-TS
!
!
interface Loopback0
  ip address 3.3.3.3 255.255.255.255
!
interface Tunnel0
  ip address 10.0.0.3 255.255.255.0
  no ip redirects
  ip mtu 1400
  ip nhrp authentication NHRPAUTH
  ip nhrp map 10.0.0.1 192.0.2.2
  ip nhrp map multicast 192.0.2.2
  ip nhrp network-id 1
  ip nhrp nhs 10.0.0.1
  ip tcp adjust-mss 1360
  tunnel source GigabitEthernet0/3
  tunnel mode gre multipoint
  tunnel protection ipsec profile DMVPN-PROFILE
!
!
interface GigabitEthernet0/3
  ip address 203.0.113.2 255.255.255.252
  duplex full
  speed 1000
  media-type rj45
!
!
router eigrp 100
  network 3.3.3.3 0.0.0.0
  network 10.0.0.0 0.0.0.255
!
router bgp 65003
  bgp log-neighbor-changes
  neighbor 203.0.113.1 remote-as 65000
```

R4 Configurations

```
hostname R4
!
crypto isakmp policy 10
  encr aes 256
  hash sha256
  authentication pre-share
  group 14
crypto isakmp key DMVPN-KEY address 0.0.0.0
!
crypto ipsec transform-set DMVPN-TS esp-aes 256 esp-sha256-hmac
mode tunnel
!
crypto ipsec profile DMVPN-PROFILE
set transform-set DMVPN-TS
!
interface Loopback0
ip address 4.4.4.4 255.255.255.255
!
interface Tunnel0
ip address 10.0.0.4 255.255.255.0
no ip redirects
ip mtu 1400
ip nhrp authentication NHRPAUTH
ip nhrp map 10.0.0.1 192.0.2.2
ip nhrp map multicast 192.0.2.2
ip nhrp network-id 1
ip nhrp nhs 10.0.0.1
ip tcp adjust-mss 1360
tunnel source GigabitEthernet0/4
tunnel mode gre multipoint
tunnel protection ipsec profile DMVPN-PROFILE
!
!
interface GigabitEthernet0/4
ip address 198.51.100.2 255.255.255.252
duplex full
speed 1000
media-type rj45
!
!
router eigrp 100
network 4.4.4.4 0.0.0.0
network 10.0.0.0 0.0.0.255
!
router bgp 65004
bgp log-neighbor-changes
neighbor 172.16.0.1 remote-as 65000
neighbor 198.51.100.1 remote-as 65000
```